The Klootwijk Architectural Paradigm

Cross-modal coordinates. Hadamard intersections. Self-referential π clocks.

Version 4.0 / The percentage-sign formulation

$$X = \pi M, \qquad Y = G \odot \mu, \qquad Z = \frac{d \cos \theta}{XY + \epsilon}$$

$$\boxed{\mathcal{T}_\pi : (N, q, \Theta) \longmapsto (N + 1, q \oplus 1, \Theta + \pi)}$$

The source matrix becomes a recorded, replayable computational object.

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Primary source: *antihistamine hatman.pdf*, 50 pages.
Mathematical specification, executable source tracker, and technical-disclosure edition.

Edition statement

V4 places the attached document's Cross-Modal Reality-Gating Framework inside the existing two-preparaat event architecture. The input channels produce the source's percentage-sign coordinates; the coordinates enter a defined state matrix; completed intervals advance one bit through π; a model-bearing record becomes input to the next interval.

The primary source develops a sequence from the Hat Man and sleep-paralysis prompts through atonia, microsaccades, IPD/IDT, MU, parameter equations, percentage-sign geometry, a tracker, and a patent-style disclosure. That sequence is retained as the conceptual basis. The executable equations and source-code rules are transcribed separately from the source's neurological interpretations and proposed application claims. [S4, pp. 1–50]

Source keys and statement types

[S4] denotes *antihistamine hatman.pdf*, the new 50-page base. **[S3]** denotes *wait I got 2 preparaats.pdf*, the 15-page preparaat and whaletail dialogue. **[S0]** denotes *trewel.pdf*, the 49-page earlier source. **[P]** denotes the author's instruction in this conversation that one tick means a one-bit flip over π.

Source equations preserve the attachment's algebra. **Derived results** follow from that algebra under stated domains. **V4 definitions** specify the extra normalization, event coupling, and finite history recurrence used in this edition's worked calculation. **Source hypotheses** retain the document's proposed neurological interpretation without turning an algebraic test into an established clinical conclusion.

The source supplies no participant dataset, receptor-calibration function, diagnostic-performance study, or intervention-outcome dataset for its thresholds. Its clinical names are preserved as source terminology and quoted program identifiers. The reference execution uses synthetic model inputs and produces numerical coordinates, records and region labels; it does not diagnose a person or command sensory treatment.

Continuity with the earlier architecture

Preparaat 1 and Preparaat 2, Alpha/Beta, hourglass occupancy, the open-cup/whaletail vocabulary, and the </> commitment/evaluation pairing remain present. Sections 20 and 21 preserve those two directions. The calculations in this edition are new executions; no byte-level equivalence with an earlier compiled V3 executable is asserted.

Reproducible artifacts

The companion package contains the evaluator, numerical model, unit tests, 48-event histories, an offline coordinate explorer, and the editable LaTeX source. The final PDF, primary source, model and evaluator are identified in an external release manifest.

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1 The cross-modal architectural statement

Basis: Source framing and memorandum: [S4, pp. 10–15].

The source names its organizing proposal the *Cross-Modal Reality-Gating Framework*. Its operational vocabulary connects sensory input, motor reference changes, a filtering coefficient, a salience weighting, and an output-coordinate assignment. In this edition, *reality-gating* is the source's name for that proposed processing model.

Let a sensory feature vector be $\boldsymbol{s}(\theta)$, a motor-reference feature be $\boldsymbol{m}$, and a collection of filter/salience values be $\boldsymbol{g}$. The architecture's basic typing is

$$(\boldsymbol{s}, \boldsymbol{m}, \boldsymbol{g}) \xrightarrow{\mathcal{F}_{\text{source}}} (\boldsymbol{E}_p, W_{\text{sal}}, X, Y, Z, J). \tag{1.1}$$

The first two outputs are the source's abstract perceptual-error and salience models. The last four are explicit scalar coordinates and a determinant. They have different mathematical types and different levels of specification.

Four operational layers

Layer	Source object	Formal role
Feature input	Eye-jitter, gating, MU, reward-anchor and drive parameters	Supplies numerical or vector-valued inputs.
Coordinate mapping	Two nodes and a diagonal slash, written as %	Computes X, Y, Z and J.
Temporal execution	Earlier π tick and Alpha/Beta registers	Advances a continuous state, then complements the bit.
Record feedback	Converging < and expanding >	Stores the completed event and reads it into the next interval.

Formal objective

The objective is to make each declared dependency inspectable: which input enters which product, which division requires a nonzero denominator, which conditional branch is selected, and which past value is read by the next update. A theorem about the resulting determinant establishes a property of that matrix. A claim about clinical states requires a separate empirical link that the attachment does not supply.

The source's application language is retained in the disclosure section as proposed use. The mathematical object specified here can be evaluated even when the clinical interpretation, physical sensor calibration, and hardware feedback plant remain unspecified. This separation permits a complete computational formulation of the supplied equations without silently inventing those missing maps.

2 Prompt lineage and the source hypotheses

Basis: Source prompts and their elaborations: [S4, pp. 1–21].

Prompt family	Source's proposed association	Retained modeling role
Antihistamine / Hat Man	Chemical disruption, filtering changes and shadow-figure interpretation.	Context for the source's exogenous-gating hypothesis.
Sleep-paralysis demon	REM-associated motor inhibition and perceived intruder/incubus states.	Source's endogenous state-intrusion category.
Atonia cat	Sleeping atonia, cataplexy, collapse and the cat-atonia word association.	Distinct source terms; no new equivalence is imposed.
Microsaccades / IPD / IDT / θ	Eye-reference changes linked to continuous binaural phase processing.	Motor feature, acoustic phase/delay interface and angle input.
Catatonia	Proposed long-lived motor/sensory feedback deadlock.	A hypothesis and legacy classifier name in the source.
Schizophrenia / MU	Gating, source monitoring, MU suppression and reward salience.	Source's proposed interpretation of low-parameter branches.
"Parameters and formalize"	Error vector, salience ratio and numerical thresholds.	Source equations and separate classifier implementations.
% / Trinity / trefoil	Two nodes, Hadamard overlap and cutting axis.	Coordinate map, matrix determinant and symbolic diagram.
"Tracker also proof it"	Numerical examples and Python tracker.	Re-executable algebra and explicitly delimited proofs.
"Formal patent" / "full disclosure"	Sensors, processing core, outputs and six draft claims.	Technical-disclosure sequence in Sections 24–26.

The attachment interleaves the author's prompts with generated responses. Author prompts identify the requested connections; response passages introduce many of the claimed mechanisms, numerical thresholds and legal-sounding assertions. This edition records that provenance rather than treating every response statement as an independently established finding.

The early medication-advice passages are not converted into architectural parameters. No dosing procedure, pharmacological intervention, or medical instruction is added to the model. The later mathematical variables are preserved as the source defines them, with their observation and calibration requirements stated separately.

3 Notation, domains and observation contracts

Basis: Parameter definitions: [S4, pp. 21–25, 29–36, 44–45]. Normalized numerical encoding: V4 definition.

Symbol	Source role	V4 numerical treatment
M	`M_jitter`, source frequency value	f_{jitter}/f_0, $f_0 = 1\,\text{Hz}$ in examples.
G	$\mathcal{G}_{p50}$ or `G_p50`	A supplied index in $[0, 1]$.
s_μ	$\Delta\mu_{\text{supp}}$ or `Mu_supp`	A supplied suppression index in $[0, 1]$.
μ	μ_{MOR}	A supplied reward-anchor index in $[0, 1]$.
d	D_2 or `D2_tone`	Nonnegative model drive.
θ	Acoustic arrival/reference azimuth	Angle in radians.
X, Y, Z	Motor node, Hadamard node, slash	Dimensionless model coordinates.
J	Source's "stability index"	Determinant of the source state matrix.
N, q, Θ	Count, bit, unwrapped event phase	$N \in \mathbb{N}_0$, $q \in \{0, 1\}$, $\Theta = N\pi$.
ϑ, ω_s	Mechanical rotor angle and spin	Separate from θ and Θ.
$c, a, b, m, X_{\text{dwell}}$	Activation, Alpha, Beta, occupancy, dwell	Inherited dynamical coordinates.

Why the unit encoding is explicit

The source calls M_{jitter} a frequency and then adds a constant 0.01 to a product containing πM_{jitter}. V4 preserves the source's numerical results by evaluating the *number* $M = f_{\text{jitter}}/(1\,\text{Hz})$. All coordinate products and $\epsilon = 0.01$ then belong to the same numerical chart. This is an added unit convention, not a claim that the attachment supplied a calibration.

Required observation map

For raw sensor channels $\boldsymbol{y}(t)$, a realizable input contract would be

$$(M, G, s_\mu, \mu, d, \theta) = C_{\text{obs}}(\boldsymbol{y}; \text{calibration record}). \tag{3.1}$$

The attachment describes possible sensors but supplies no complete C_{obs}. In particular, a parameter named after a receptor is not made a measured receptor quantity by inserting it into the matrix. The synthetic reference cases bypass this missing map by declaring their numbers as model inputs.

The MU suppression index, MOR parameter, fluid viscosity μ_f and chemical potential μ_i remain distinct. Likewise the gating coefficient G is not the gravity vector $\boldsymbol{g}$, and motor-coordinate X is not elapsed dwell X_{dwell}.

4 Microsaccades, IPD, IDT and the angle interface

Basis: The auditory/oculomotor discussion: [S4, pp. 7–14]; literal phase update: [S4, p. 26].

The source gives IDT two readings: interaural delay time and interdigital transducer. Its later equations use $\mathcal{H}_{IDT}$ as a delay/reference operator. The transducer interpretation is not used by the displayed tracker and remains an alternative source term.

For a single sinusoidal component and a chosen left/right sign convention, the phase–delay identity is

$$\Delta\phi_{LR} = 2\pi f \Delta t_{LR} \pmod{2\pi}. \tag{4.1}$$

It follows directly by subtracting phases of two time-shifted sinusoids. This identity does not specify the map from eye movement to ear delay, or the inverse map from measured delay to azimuth. The attachment proposes such a cross-modal relation but supplies neither a head/ear geometry model nor a fitted transfer function.

Literal source-code phase update

The C++ routine applies

$$\theta' = \begin{cases} \theta - 0.05G \sin(\delta_{\text{eye}} M), & M > 0.1, \\ \theta, & M \le 0.1. \end{cases} \tag{4.2}$$

Here δ_{eye} is the code-valued `eyeJitterAngle`. The normalized M convention in Section 3 makes the trigonometric argument numerically defined. The source's literal coefficients and strict inequality are preserved. The equation is an update rule for a model angle; no experimentally established physiological correction gain is inferred from 0.05.

An unresolved PLL specification

A complete phase-locked loop would have to identify its reference, error signal, controlled state and update or transfer law. Equation (4.2) specifies one angle update. It does not specify a delay-line plant, an auditory error measurement or a measured closed-loop response. V4 therefore stores this routine as a separate source function, not as proof of the source's claimed cross-modal mechanism.

Three angles, three meanings

The external coordinate θ enters $\cos\theta$ in the slash equation. The architectural event phase Θ counts π-advances. The mechanical orientation ϑ rotates material tensors. Section 19 introduces a separate, explicitly encoded $\theta + \pi q$ view; that representation does not silently change the physical sound-arrival angle.

5 The perceptual-error equation

Basis: Exact source equation and definitions: [S4, pp. 21–22].

The source's vector expression is

$$\boxed{\boldsymbol{E}_p(\theta) = \frac{1}{G}\boldsymbol{S}_{\text{noise}}(\theta) + (1 - s_\mu)\boldsymbol{T}_{\text{biomotion}} - \boldsymbol{M}_f(\delta_{\text{eye}})\,\mathcal{H}_{IDT}.} \tag{5.1}$$

The three terms are the source's noise-amplification component, unmirrored-motion component and motor/delay correction component. The signs and placements are unchanged.

Minimal mathematical typing

Choose a feature space $V = \mathbb{R}^k$ and require all three terms to lie in V. A compatible interpretation is

$$\boldsymbol{S}_{\text{noise}}, \boldsymbol{T}_{\text{biomotion}} \in V, \qquad \mathcal{H}_{IDT} \in U, \qquad \boldsymbol{M}_f(\delta_{\text{eye}}) \in \mathcal{L}(U, V). \tag{5.2}$$

This is a typing of the displayed product. The source does not supply the feature dimension, the entries of $\boldsymbol{M}_f$, or the delay feature in U. It is consequently a formal dependency equation, not a fully parameterized numerical subsystem of the supplied code.

Domain of the displayed reciprocal

Although the source lists $G \in [0, 1]$, its reciprocal is defined only for $G > 0$. For nonzero fixed $\boldsymbol{S}_{\text{noise}}$,

$$\left\|G^{-1}\boldsymbol{S}_{\text{noise}}\right\| = G^{-1}\|\boldsymbol{S}_{\text{noise}}\| \longrightarrow \infty \quad (G \downarrow 0). \tag{5.3}$$

No replacement $G \mapsto \max(G, \delta)$ is inserted. At $G = 0$, Equation (5.1) is recorded as undefined unless a separate source-supported extension is provided. The percent-coordinate equation later uses a different regularizer, $XY + \epsilon$, and remains finite on its declared nonnegative domain.

Mathematical effect versus interpretation

For fixed vectors, reducing s_μ increases the scalar coefficient of $\boldsymbol{T}_{\text{biomotion}}$. That algebra alone does not determine the norm or direction of the full sum, because the vectors may reinforce or cancel. The source's phrases about decoded intent, threat projections and clinical states are interpretation hypotheses attached to this expression; they are not consequences of vector addition alone.

The reference tracker evaluates the explicitly supplied scalar map. It does not invent the missing $\boldsymbol{S}_{\text{noise}}$, $\boldsymbol{T}_{\text{biomotion}}$ or $\boldsymbol{M}_f\mathcal{H}_{IDT}$ arrays.

6 Salience weighting and the C++ specialization

Basis: General salience equation: [S4, p. 22]; implemented denominator: [S4, p. 25].

The source writes

$$W_{\text{sal}} = \frac{\alpha d}{\beta\mu + \gamma g_A}, \tag{6.1}$$

where g_A is its GABA-A control-tone parameter and α, β, γ are scaling constants. The denominator must be nonzero for the displayed ratio to be defined. With positive coefficients and nonnegative terms, the ratio is nonnegative.

The C++ code then uses a particular, narrower expression:

$$W_{\text{cpp}} = \begin{cases} d/(2.5\mu + 1.5), & 2.5\mu + 1.5 > 0, \\ 999, & 2.5\mu + 1.5 \le 0. \end{cases} \tag{6.2}$$

Its struct contains no g_A input. Relative to Equation (6.1), this corresponds numerically to $\alpha = 1$, $\beta = 2.5$ and the fixed product $\gamma g_A = 1.5$. The source does not separately identify the values of γ and g_A.

Consequences in the source parameter domain

For $0 \le \mu \le 1$,

$$1.5 \le 2.5\mu + 1.5 \le 4, \qquad \frac{d}{4} \le W_{\text{cpp}} \le \frac{d}{1.5} \quad (d \ge 0). \tag{6.3}$$

The fallback value 999 is unreachable within that domain; it is retained in the provenance implementation because it is present in the source code.

For $d > 0$,

$$\frac{\partial W_{\text{cpp}}}{\partial d} = \frac{1}{2.5\mu + 1.5} > 0, \qquad \frac{\partial W_{\text{cpp}}}{\partial \mu} = -\frac{2.5d}{(2.5\mu + 1.5)^2} < 0. \tag{6.4}$$

These statements describe the selected ratio. They do not identify an unknown person's receptor levels from an EEG channel or from a sound stimulus.

Two models remain two models

The percent tracker does not use W_{cpp} in its determinant. It inserts d directly into Z and the lower-right matrix entry. The salience classifier and determinant classifier are therefore distinct routines, even though the source gives them overlapping clinical-sounding names. V4 does not merge their thresholds into one hidden decision rule.

7 The source parameter table and its thresholds

Basis: Source table reproduced as model specifications: [S4, p. 23].

The numerical intervals below are the attachment's entries. The source calls the first numeric column "Healthy Range" and the second "Psychosis Threshold." V4 preserves the values as *source-proposed regimes*; the attachment supplies no calibration or diagnostic-performance data establishing those column labels.

Parameter	Source nominal range	Source alternate entry	Computational role
G	0.75–1.00	< 0.30	Filter index and Hadamard factor.
s_μ	0.60–0.90	< 0.15	C++ classifier condition; not an entry of $\boldsymbol{M}_{\text{state}}$.
M_{jitter}	0.1–2.0 Hz	0 Hz or "erratic"	Source numerical motor coordinate.
μ	0.70–1.00	< 0.25	Hadamard factor and salience denominator.
d	1.0–2.5	> 5.0	Source drive and slash multiplier.

Thresholds are implementation-specific

The C++ state machine uses additional thresholds $G < 0.20$, $G < 0.15$, $M \leq 0.1$, and salience cutoffs 2.0, 3.5, 4.5 and 5.0. The Python tracker instead uses $J \leq 0$, $J < 2$ and $M \leq 0.1$. The separate overlap test uses $|X - Y| < 0.15$ and $|Z| > 4.5$. All three are listed explicitly in Section 12.

A table range, a classifier predicate and a matrix sign boundary are three different objects. None is substituted for another. For example, the source's $d > 5$ entry does not by itself imply that J becomes negative: for $d > 0$, d factors out of the sign calculation.

Parameter provenance record

For an implemented observation channel, a useful typed record is

$$\mathsf{P}_i = (\text{name}, \text{value}, \text{unit}, \text{origin}, \text{timestamp}, \text{calibration ID}). \tag{7.1}$$

This is a V4 data-contract definition. In the supplied examples, `origin` is `synthetic/source-example`. Unknown biological availability is not silently replaced by the nominal interval. No user-specific medical value is inferred in this edition.

8 The percentage sign and the two Hadamard nodes

Basis: Author's percent-sign prompt and the source decomposition: [S4, pp. 26–30].

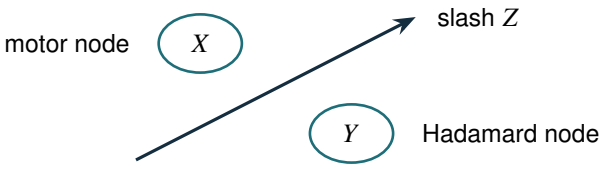
The author describes two balls as Hadamard products and the cutting slash as a third overlapping axis. The subsequent response gives the right node an explicit product, $G \odot \mu$, but implements the left node as πM. These two levels of description are preserved rather than silently treating the left node as an unspecified additional product.

$$X = \pi M, \qquad Y = G \odot \mu. \tag{8.1}$$

For scalars, $G \odot \mu = G\mu$. For equal-length vectors,

$$\boldsymbol{Y} = \boldsymbol{G} \odot \boldsymbol{\mu}, \qquad Y_i = G_i \mu_i. \tag{8.2}$$

If two separate product-valued balls are desired, the author's broader phrase can be typed as $\boldsymbol{A} = \boldsymbol{a}_1 \odot \boldsymbol{a}_2$ and $\boldsymbol{B} = \boldsymbol{b}_1 \odot \boldsymbol{b}_2$; the attachment does not specify a separate left-node factor pair. The executable branch remains Equation (8.1).



Symbolic two-node/slash organization from the source prompt, not a plot of the numerical coordinate map.

A three-object data geometry

The symbol % organizes two nodes and one cross-cutting coordinate. Define the source map

$$\mathcal{P}_{\%} : (M, G, \mu, d, \theta) \mapsto (X, Y, Z). \tag{8.3}$$

The third coordinate is a rational function of the first two, drive and angle; it is not an independently measured line slope. The source's later 45-degree drawing language specifies a visual arrangement, not an additional equality $\theta = \pi/4$ for every input.

Relation to the older double Hadamard encounter

The earlier architecture uses

$$C = W \odot (P_1 \odot P_2). \tag{S3}$$

The new product $G \odot \mu$ and the old encounter $P_1 \odot P_2$ share an elementwise operation, but their operands have different meanings. They are linked through a declared feature or drive map, not by asserting $G = P_1$ or $\mu = P_2$. Section 22 declares the actual numerical feature map used in the V4 run.

The retained construction is two source nodes, a coupled slash, an explicit state matrix, and a time-indexed record. The word *Trinity* names the three-part source organization; its algebra is stated independently of any topological claim.

9 The exact coordinate map and state matrix

Basis: Source matrix and Python implementation: [S4, pp. 30–36, 45].

For $M, d \geq 0$, $G, \mu \in [0, 1]$, $\theta \in \mathbb{R}$, and $\epsilon > 0$, define

$$X = \pi M, \qquad Y = G\mu, \qquad \boxed{Z = \frac{d \cos\theta}{XY + \epsilon}.} \tag{9.1}$$

The matrix and index are

$$\boldsymbol{M}_{\text{state}} = \begin{pmatrix} X & Y \\ Z & d \end{pmatrix}, \qquad \boxed{J = \det \boldsymbol{M}_{\text{state}} = Xd - YZ.} \tag{9.2}$$

The source's regularizer is $\epsilon = 0.01$ in all three published numerical examples. The denominator is at least ϵ in the declared domain.

Scalar, not block, determinant

Although the source uses bold symbols, its executable values are scalars. A determinant of a matrix of vector-valued blocks is not supplied. For an array implementation, V4 can evaluate independent channels,

$$\boldsymbol{M}_i = \begin{pmatrix} \pi M_i & G_i\mu_i \\ d_i \cos\theta_i / (\pi M_i G_i \mu_i + \epsilon) & d_i \end{pmatrix}, \qquad J_i = \det \boldsymbol{M}_i. \tag{9.3}$$

This is a channelwise extension, not a replacement of the source scalar rule. The reference calculations below use one scalar channel.

Exact dependence on the angle

For fixed X, Y, d, ϵ,

$$Z(\theta + 2\pi) = Z(\theta), \qquad Z(\theta + \pi) = -Z(\theta), \tag{9.4}$$

$$J(\theta) + J(\theta + \pi) = 2Xd. \tag{9.5}$$

This mathematical half-turn identity will connect the slash to the bit's encoded orientation. It does not mean that an atomic carrier half-period is automatically an acoustic arrival-angle change.

What the determinant certifies

For this 2×2 real matrix, $J = 0$ is precisely the singular-matrix condition. It determines invertibility and oriented area scaling of the linear map represented by $\boldsymbol{M}_{\text{state}}$. A claim about temporal stability additionally requires a dynamical law; no such law is specified by the determinant alone. Section 10 derives the actual zero set without renaming it a clinical boundary.

10 The determinant boundary: exact derivation

Basis: Derivation from the source equations in Section 9.

Substitute the source expression for Z:

$$J = d\left[X - \frac{Y\cos\theta}{XY+\epsilon}\right] \tag{10.1}$$

$$= d\frac{X^2Y + \epsilon X - Y\cos\theta}{XY+\epsilon}. \tag{10.2}$$

For $d > 0$ and the nonnegative input domain, the denominator is positive. Therefore

$$\boxed{\operatorname{sgn} J = \operatorname{sgn}\left(X^2Y + \epsilon X - Y\cos\theta\right).} \tag{10.3}$$

This is the sign criterion implied by the attachment's equations.

Positive-angle-cosine branch

For $Y > 0$ and $\cos\theta > 0$, the unique nonnegative root is

$$X_c = \frac{-\epsilon + \sqrt{\epsilon^2 + 4Y^2\cos\theta}}{2Y} = \frac{2Y\cos\theta}{\epsilon + \sqrt{\epsilon^2 + 4Y^2\cos\theta}}. \tag{10.4}$$

The second expression avoids subtracting nearly equal positive quantities. Since the polynomial derivative is $2XY + \epsilon > 0$,

$$J < 0\ (0 \le X < X_c), \qquad J = 0\ (X = X_c), \qquad J > 0\ (X > X_c). \tag{10.5}$$

The motor-coordinate boundary is $M_c = X_c/\pi$. At $\theta = 0$ and $\epsilon = 0.01$, the source example products give $M_c \approx 0.316236228$ for $Y = 0.765$ and $M_c \approx 0.269648927$ for $Y = 0.030$.

Other domain branches

If $Y = 0$, then $J = dX$. If $d = 0$, both Z and J vanish. If $\cos\theta \le 0$, the numerator in Equation (10.3) is nonnegative; it is strictly positive for $X > 0$ and $d > 0$. These are branches of one algebraic surface, not separate diagnostic rules.

A precise proof boundary

The derived result proves the location of $J = 0$ in this model. It does not prove that the slash stops moving there: the source matrix contains no equation for $\dot{\theta}$ or $\dot{Z}$. It also does not prove a knot-type change, because no embedded closed curve or evolving knot family is defined by this matrix.

11 Sensitivity, regularization and limiting cases

Basis: Analytical consequences of the unchanged source formula.

For fixed d, θ, ϵ,

$$\frac{\partial J}{\partial X} = d\left[1 + \frac{Y^2 \cos\theta}{(XY+\epsilon)^2}\right], \tag{11.1}$$

$$\frac{\partial J}{\partial Y} = -\frac{d\epsilon \cos\theta}{(XY+\epsilon)^2}, \tag{11.2}$$

$$\frac{\partial J}{\partial \theta} = \frac{dY \sin\theta}{XY+\epsilon}. \tag{11.3}$$

When $d > 0$ and $\cos\theta \geq 0$, J increases with X and does not increase with Y. Hence *reducing the Hadamard filter product increases* J in this branch. Meanwhile

$$\frac{\partial Z}{\partial Y} = -\frac{dX \cos\theta}{(XY+\epsilon)^2}, \tag{11.4}$$

so lowering Y can increase Z while also increasing J. The factor Y in the subtractive term YZ matters.

The source's prose says a drop in G or μ necessarily drives the determinant into a collapse region [S4, pp. 45–47]. That monotonic claim does not follow from the displayed matrix. V4 preserves the matrix and records this difference explicitly rather than modifying the denominator or sign to fit the prose.

Finite regularized values

For $X, Y \geq 0$,

$$|Z| \leq d/\epsilon, \qquad Z\big|_{X=0} = d\cos\theta/\epsilon, \qquad \lim_{Y\downarrow 0} J = dX. \tag{11.5}$$

The slash is rational, not exponential, in the product XY. With fixed positive ϵ, the model has no denominator singularity on this domain. It can still have a large numerical slash value.

Drive magnitude and actual time stability

For $d > 0$, the sign of J is independent of the magnitude of d. Multiplying d multiplies J and Z together. It can change whether $J \geq 2$, but not the sign at a fixed (X, Y, θ).

A positive determinant by itself is not a temporal-stability proof: the invented test system $\dot{\boldsymbol{z}} = \boldsymbol{I}\boldsymbol{z}$ grows, whereas $\dot{\boldsymbol{z}} = -\boldsymbol{I}\boldsymbol{z}$ decays, and both matrices have determinant 1. The source calls $\boldsymbol{M}_{\text{state}}$ a state representation, not a fully specified differential-equation generator. Its numerical label is therefore retained as an index name, not a substitute for a stability theorem.

12 Three distinct source classifiers

Basis: C++ state machine, overlap test and Python tracker: [S4, pp. 24–26, 30–35].

A. C++ salience classifier: ordered branches

Let $w = d/(2.5\mu + 1.5)$ within the declared parameter domain. When `isAwake` is true, the first matching row is returned:

Condition, in source order	Literal source identifier
$G < 0.30$, $s_\mu < 0.15$, $w > 3.5$	`SCHIZOPHRENIC_GATING_COLLAPSE`
$M \leq 0.1$, $G < 0.20$, $w > 4.5$	`CATATONIC_DEADLOCK`
$G < 0.15$, $w > 5.0$	`ANTICHOLINERGIC_DELIRIUM`
Otherwise	`STABLE_COHERENCE`

When `isAwake` is false, the source returns `REM_INTRUSION_PARALYSIS` if `remPhaseIntrusion` is true and $w > 2$; otherwise it returns `STABLE_COHERENCE`. This is the literal Boolean ordering, not a new physiological interpretation of the input flags.

B. C++ coordinate-overlap flag

The later topology routine computes $X = 3.14159M$, $Y = G\mu$, and $Z = d\cos\theta/(XY + 0.01)$, then returns

$$B_{\text{overlap}} = \mathbf{1}_{\{|X-Y|<0.15 \,\wedge\, |Z|>4.5\}}. \tag{12.1}$$

This test uses no determinant. The literal C++ approximation 3.14159 also differs from the Python library value of π. Both evaluation choices are available in the V4 reference functions.

C. Python determinant classifier

The source Python tracker evaluates the unrounded determinant before selecting

$$\mathcal{R}_J = \begin{cases} \texttt{NONPOSITIVE_DETERMINANT}, & J \leq 0, \\ \texttt{POSITIVE_BELOW_TWO}, & 0 < J < 2, \\ \texttt{AT_LEAST_TWO}, & J \geq 2. \end{cases} \tag{12.2}$$

These V4 output names describe exactly the numerical regions. The corresponding original strings are retained in the `source_python_label` field and listed on the next page.

Python source identifiers, preserved as identifiers

Source condition	Literal source label
$J \leq 0$, $M \leq 0.1$	`CATATONIC_DEADLOCK (Trefoil Phase Lock)`
$J \leq 0$, $M > 0.1$	`ACUTE_GATING_COLLAPSE (Psychosis State)`
$0 < J < 2$	`PRODROMAL_SENSORY_LEAKAGE (High Vulnerability)`
$J \geq 2$	`STABLE_REALITY_COHERENCE (Operational Baseline)`

The executable V4 record stores these strings as source provenance. Its human-readable coordinate explorer uses Equation (12.2). No patient-specific diagnosis or receptor estimate is generated from a synthetic model number.

Why the branches cannot be silently combined

The C++ routine can satisfy multiple predicates at once. For example, with $G = 0.1$, $s_\mu = 0.1$, $M = 0.05$, $\mu = 0.1$, $d = 20$ and `isAwake=true`, all three non-default salience conditions hold, but the first one wins. Increasing only s_μ to 0.2 bypasses the first condition and returns the second.

The overlap test and determinant test can also disagree. The source's third input example, $M = 0.08$, $G = 0.12$, $\mu = 0.18$, $d = 7.2$, $\theta = 0$, has $J < 0$ but $|X - Y| > 0.15$. It therefore returns a nonpositive determinant region while the overlap flag is false.

Equality conventions

The Python code sends $J = 2$ to its final branch because the intermediate comparison is strict. Some later source prose uses $J > 2$ for the nominal region. V4's executable preserves the code's $J \geq 2$ behavior and records the prose/code difference. Likewise $M = 0.1$ enters the low-jitter branch in the Python classifier, while the phase-adjustment routine activates only when $M > 0.1$.

> All three rule sets can be reproduced. They are not equivalent tests, and reproducing any one of them is not a proof that its clinical-sounding label is a validated diagnosis.

13 Source figure, Venn language and trefoil meaning

Basis: Source diagram on page 29 and geometric language on pages 26–31.

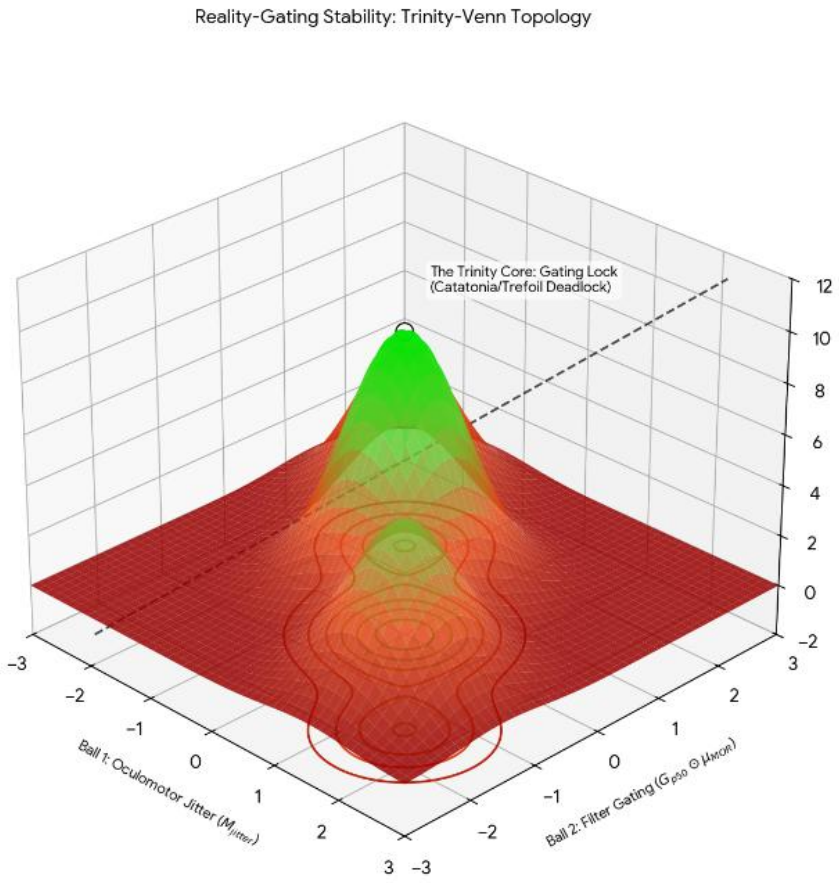


Figure 1. Original embedded figure from [S4, p. 29]. Its title is "Reality-Gating Stability: Trinity-Venn Topology." It shows a peaked colored surface, contours and a dashed diagonal near a labeled central region.

The attachment provides no mesh-generation equation or plotted dataset for this image. Its axes extend into negative coordinate values, whereas the declared scalar input domain has $X, Y \geq 0$. V4 retains the figure as a source diagram; it does not present the colored surface as a numerical rendering of Equation (9.1).

The percent sign, Venn intersection and trefoil are three different descriptions in the dialogue. A Venn construction would require specified sets and their intersections. A knot construction would require a specified closed spatial embedding. The source supplies neither through the 2×2 determinant. Its implemented object is the coordinate matrix and the separate overlap predicate. No missing surface or knot formula is invented here.

14 The source-faithful numerical tracker

Basis: Python tracker and example code: [S4, pp. 33–37].

The computational core evaluates six scalar inputs $(M, G, \mu, d, \theta, \epsilon)$ and returns coordinates, determinant, region, literal source label and overlap flag. It does not round before a comparison.

```
x = pi * M_jitter
y = G_p50 * mu_MOR
z = D2_tone * cos(theta) / (x*y + epsilon)
J = x*D2_tone - y*z
if J <= 0:
    region = "NONPOSITIVE_DETERMINANT"
elif J < 2:
    region = "POSITIVE_BELOW_TWO"
else:
    region = "AT_LEAST_TWO"
```

The source rounds returned coordinates and the index to three decimals. V4 exports full floating-point values and rounds only its printed tables. This changes presentation precision, not the underlying source formula or branch ordering.

Implemented functions

Function	Role
`coordinates`	Source scalar map and Python region logic; configurable π value.
`source_cpp_label`	Exact ordered C++ salience classifier.
`source_cpp_phase`	Literal angle-update formula.
`critical_x`	Analytical zero boundary from Section 10.
`simulate`	Declared V4 coupling to the inherited clock and record system.
`verify`	Offline algebraic and replay checks on emitted rows.

Input contract and output contract

Finite numerical inputs are required; $M, d \geq 0$, $G, \mu \in [0, 1]$ and $\epsilon > 0$ define the evaluated branch. Values outside that domain cause an input error rather than an invented clinical classification. This is the mathematical domain of the implementation, not an added behavioral intervention.

The tracker is a calculation from supplied numbers. “Real time” would additionally require live observation functions, timestamp handling, buffer scheduling and an output plant. The source describes those interfaces but does not supply their complete implementations. The companion offline explorer makes the numerical dependency interactive without simulating a biological sensor.

15 Re-executed source examples

Basis: Examples A and B: [S4, pp. 32–33, 35–37]. Example C: [S4, p. 35].

The source's example inputs are retained, with $\theta = 0$ and $\epsilon = 0.01$. Example names A, B and C below are neutral record keys; the attachment assigns clinical-sounding names to some of them.

Example	M	G	μ	d
A	1.50	0.85	0.90	1.50
B	0.05	0.15	0.20	8.50
C	0.08	0.12	0.18	7.20

Example	X	Y	Z	J
A	4.712388980	0.765000	0.414940334	6.751154115
B	0.157079633	0.030000	577.744376616	-15.997154421
C	0.251327412	0.021600	466.663621523	-8.270376856

The values in the source's displayed A and B summaries agree with the calculations when rounded to three decimals: $(4.712, 0.765, 0.415, 6.751)$ and $(0.157, 0.030, 577.744, -15.997)$. Example C follows directly from the source's separate tracker call.

A numerical distinction already present in the source

For example C, $|X - Y| \approx 0.229727 > 0.15$ and the overlap flag is false, while $J \approx -8.270377 < 0$. Thus the printed algorithm variants give different outputs for a source-provided input. V4 preserves that difference instead of claiming a single validated "trefoil collapse" criterion.

What the execution establishes

It establishes that the three input tuples produce the listed numerical values under the published formula. It also establishes the analytical sign test, the finite regularized boundary and the classifier precedence when implemented as written. It does not supply clinical observations, outcome prediction accuracy, or proof that the source's feedback signals restore motor or perceptual function.

Determinant sweep from the actual formula

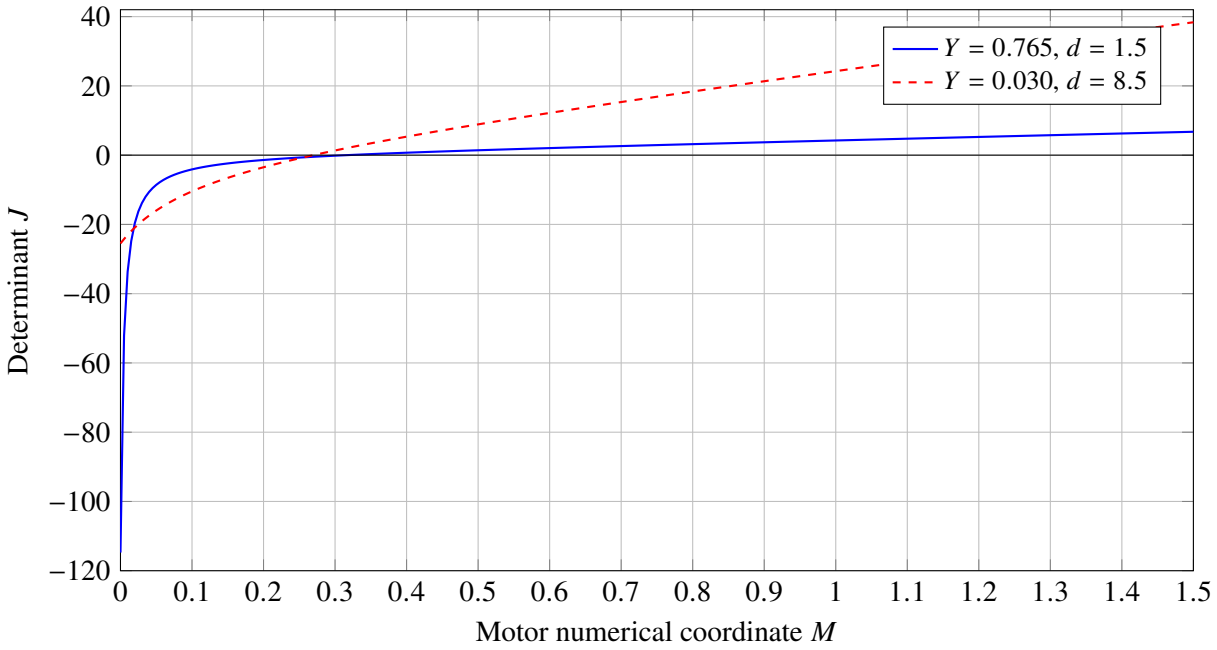


Figure 2. Newly executed mathematical sweep, with $\theta = 0$ and $\epsilon = 0.01$. Each curve changes only M and holds its stated Y, d values. The plotted sign crossings agree with Equation (10.4).

At $M = 0$, the two initial values are -114.75 and -25.5 respectively. Both are shown, and the complete 301-row sweep is supplied in `determinant_sweep.csv`. The same formula is used at the low-motor endpoint and throughout the plotted domain.

The model therefore exposes two different operations: altering a numerical coordinate and altering a biological interpretation attached to that coordinate. Only the first is executed by this sweep.

16 Preparaat conversion and the biochemical clock

Basis: Conserved preparaat roles and rate modulation: [S3, pp. 3–4, 12]; thermal rate: [S0, pp. 23–25].

Retain the selected convertible population

$$P_1 = P_\star(1 - c), \qquad P_2 = P_\star c, \qquad P_1 + P_2 = P_\star, \tag{16.1}$$

with the reversible interval law

$$\dot{c} = k_{01}(1 - c) - k_{10}c. \tag{16.2}$$

For held positive rates, its exact solution over one interval is

$$c_\infty = \frac{k_{01}}{k_{01} + k_{10}}, \qquad c' = c_\infty + (c - c_\infty)e^{-(k_{01}+k_{10})\Delta t}. \tag{16.3}$$

This is a convex combination of c and c_∞, preserving the selected population interval $[0, 1]$.

Chemical time and thermal dependence

The inherited Arrhenius branch and its reference-temperature form are

$$k(T) = Ae^{-E_a/(RT)}, \qquad k_T = k_r \exp\left[\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T}\right)\right]. \tag{16.4}$$

For the irreversible branch with normalized exposure $i(t)$,

$$\zeta(t) = \int_0^t k(T(s))i(s)\,\mathrm{d}s, \qquad c(t) = 1 - [1 - c(0)]e^{-\zeta(t)}. \tag{16.5}$$

The progress coordinate ζ is not the atomic carrier phase or the sensory azimuth. A biochemical realization must identify its switch population and rates; no equation supplied in [S4] makes G, μ or d identical to that population.

Source rate and declared V4 reverse branch

The earlier preparaat source displays

$$k_{01,n} = k_T \frac{1 + 0.35a_n + 0.25\bar{h}_n}{1 + 0.5m_n}. \tag{16.6}$$

The worked V4 instance selects

$$k_{10,n} = 0.6k_T \frac{1 + 0.2b_n}{1 + 0.5m_n}. \tag{16.7}$$

The reverse coefficient and numerical time constants are stated as instance choices in Section 22. They are not receptor kinetics inferred from the new source's neurological labels. The new feature drives the same typed continuous state rather than replacing its conservation law.

17 Atomic reference and the one-bit half-turn

Basis: Inherited tick instruction [P] and event operator [S3, p. 4]; external metrology reference [E1].

For atomic states with energy difference ΔE,

$$\Delta E = h\nu_A = \hbar\omega_A, \qquad \omega_A = 2\pi\nu_A. \tag{17.1}$$

The selected reference frequency is the defining caesium-133 value [E1],

$$\nu_{\mathrm{ref}} = 9\,192\,631\,770\,\mathrm{Hz}. \tag{17.2}$$

This is the metrological reference channel. A numerical example using that frequency is not itself a physical atomic-clock experiment.

Architectural event

$$\boxed{\mathcal{T}_\pi : (N, q, \Theta) \mapsto (N + 1, q \oplus 1, \Theta + \pi).} \tag{17.3}$$

For $N_0 = q_0 = \Theta_0 = 0$, $q_n = n \bmod 2$ and $\Theta_n = n\pi$. The circular phase $e^{i\Theta_n}$ changes sign on each event. The unwrapped phase and record count continue increasing when the bit returns after two events.

For a mathematical two-level drive about the Pauli x axis,

$$U(\beta) = e^{-i\beta\sigma_x/2}, \qquad U(\pi) = -i\sigma_x, \qquad U(\pi)|q\rangle = -i|1 - q\rangle. \tag{17.4}$$

Here β is the drive's pulse area. This algebraic inversion is a possible two-state realization, distinct from advancing an unforced carrier by half a period.

Integer cycle divider

For intended event rate $r_n > 0$,

$$D_n = \left\lceil \frac{\nu_{\mathrm{ref}}}{r_n} \right\rceil, \quad \Delta t_n = D_n/\nu_{\mathrm{ref}}, \quad C_{n+1} = C_n + D_n. \tag{17.5}$$

During the interval, define

$$\Theta(t) = n\pi + \frac{\Phi_A(t) - \Phi_A(t_n)}{2D_n}. \tag{17.6}$$

Then a completed D_n-cycle interval advances Θ by exactly π, and

$$0 \le \Delta t_n - 1/r_n < 1/\nu_{\mathrm{ref}}. \tag{17.7}$$

The event stores the integer $\Theta_n/\pi = n$; it does not double-count an additional jump after a phase endpoint already reached during continuous evolution.

18 Alpha, Beta, hourglass and retained history

Basis: Alpha/Beta and fill state [S3, p. 4]; inverse-log history and recovery [S0, pp. 29–30].

The inherited coupled temporal states are

$$\tau_\alpha \dot{a} = -a + \chi b + u, \qquad \tau_\beta \dot{b} = a - b, \qquad 0 < \tau_\alpha < \tau_\beta, \quad 0 \le \chi < 1. \tag{18.1}$$

For a held input, let

$$\boldsymbol{A} = \begin{pmatrix} -1/\tau_\alpha & \chi/\tau_\alpha \\ 1/\tau_\beta & -1/\tau_\beta \end{pmatrix}, \qquad \boldsymbol{y}_\infty = \frac{u}{1-\chi}\begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{18.2}$$

Then

$$\boldsymbol{y}' = \boldsymbol{y}_\infty + e^{\boldsymbol{A}\Delta t}(\boldsymbol{y} - \boldsymbol{y}_\infty). \tag{18.3}$$

The trace of $\boldsymbol{A}$ is negative and its determinant is $(1-\chi)/(\tau_\alpha\tau_\beta) > 0$, establishing decay of this held-input linear subsystem. This is a proper dynamical stability statement about $\boldsymbol{A}$, not about the source's coordinate matrix $\boldsymbol{M}_{\text{state}}$.

Hourglass branch

During a held bit interval, the selected cyclic reservoir update is

$$m' = \begin{cases} 1 - (1-m)e^{-\eta\Delta t}, & q = 1, \\ \max\{0, (1+m)e^{-\Delta t/\tau_r} - 1\}, & q = 0. \end{cases} \tag{18.4}$$

It corresponds to filling at $\dot{m} = \eta(1-m)$ and returning at $\dot{m} = -(1+m)/\tau_r$ until the reservoir is empty. The return time from m_0 is $\tau_r \ln(1+m_0) \le \tau_r \ln 2$. Dwell accumulates as $X'_{\text{dwell}} = X_{\text{dwell}} + q\Delta t$.

An archive and a readout of that archive

For retained feature values v_i, define the V4-selected normalized readout

$$\bar{h}_n = \frac{\sum_{i=1}^{n} w_{n-i} v_i}{\sum_{i=1}^{n} w_{n-i}}, \qquad w_j = \frac{1}{\ln(j+e)}, \qquad \bar{h}_0 = 0. \tag{18.5}$$

The earlier source supplies the inverse-log weighting idea. V4 specifies normalization and the feature being stored. With $v_i \in [0,1]$, the readout remains in $[0,1]$. The original records remain unchanged while the weighted readout changes with event age.

19 Whaletail continuity and encoded slash orientation

Basis: Open cups, whaletail and quarter-turn conductance [S3, pp. 10–15]; new percent/slash link is a V4 definition.

The inherited material geometry distinguishes open cup surfaces, a central interface, external stick-poles and a drive path. Its whaletail description names a two-wing routing domain. These are spatial structures; the new (X, Y, Z) tuple is a feature-state coordinate. V4 links them through separate maps rather than assigning the same units to both.

The bit-selected mechanical angle and conductance remain

$$\varphi_q = \frac{\pi}{2} q, \qquad \boldsymbol{R}_q = \boldsymbol{R}_z(\varphi_q), \qquad \boldsymbol{K}_q = \boldsymbol{R}_q \boldsymbol{K}_0 \boldsymbol{R}_q^{\mathsf{T}}. \tag{19.1}$$

With rotor angle ϑ, use $\boldsymbol{Q} = \boldsymbol{R}_z(\vartheta)\boldsymbol{R}_q$. For planar principal values $\kappa_\parallel, \kappa_\perp$ and $\alpha = \vartheta + \pi q/2$,

$$K_{xx} = \kappa_\parallel \cos^2 \alpha + \kappa_\perp \sin^2 \alpha, \qquad K_{xy} = (\kappa_\parallel - \kappa_\perp) \sin \alpha \cos \alpha. \tag{19.2}$$

The double-angle encoding $2\varphi_q = \pi q$ connects a quarter-turn director state to a binary half-turn.

The percent slash as an encoded view

Define, separately from the source angle,

$$Z_{\mathrm{enc}}(q) = (-1)^q Z(\theta) = Z(\theta + \pi q), \qquad J_{\mathrm{enc}} = Xd - YZ_{\mathrm{enc}}. \tag{19.3}$$

One bit complement reverses the *encoded* slash. For a fixed input tuple, its two views satisfy $J_{\mathrm{enc}}(0) + J_{\mathrm{enc}}(1) = 2Xd$. The original $J(\theta)$ remains in the record and continues to determine the source region. The model cannot produce a clinical improvement merely by changing its display orientation.

Typed spatial transfer

A device-specific map could send a feature output to a torque, a pressure boundary, or a rendering parameter:

$$\mathcal{B} : (X, Y, Z, \bar{h}, q) \mapsto (\tau_s, p_{\Gamma_0}, p_{\Gamma_H}, \boldsymbol{u}_{\mathrm{render}}). \tag{19.4}$$

The attachment does not supply that full map. The reference instance chooses only a normalized spin target and numerical drive term; it does not solve a whaletail pressure field or command the source's proposed therapeutic feedback.

20 Converging commitment: the < record

Basis: Section-20 lineage and model-bearing records: [S3, pp. 6–9]; digest reference [E2].

After a continuous interval ends, the converging operation gathers its input tuple, source-coordinate result, clock endpoint and model state into a finite payload. In an optical realization, the older commitment is

$$E_{\mathrm{exp},n}(\boldsymbol{x}) = \int_{t_n}^{t_{n+1}} I_{\mathrm{sensor}}(\boldsymbol{x}, t)\,\mathrm{d}t, \qquad \mathsf{D}_{n+1}^{\mathrm{optical}} = \mathcal{QR}(E_{\mathrm{exp},n}). \tag{20.1}$$

The V4 numerical realization uses the same commitment role for a structured event record.

Source-bound identities

Let h_S identify the primary source PDF bytes, h_E the evaluator bytes and $h_{\mathcal{M}}$ the canonical model description:

$$h_S = \mathrm{SHA256}(S), \qquad h_E = \mathrm{SHA256}(E), \qquad h_{\mathcal{M}} = \mathrm{SHA256}(\mathrm{enc}\,\mathcal{M}). \tag{20.2}$$

SHA-256 is the specified message-digest algorithm [E2]. The full payload is retained separately from its 256-bit digest.

The event payload

$$\mathsf{P}_{n+1} = \big(h_S, h_E, h_{\mathcal{M}}, h_n, N_{n+1}, q_n, q_{n+1}, C_{n+1}, D_n, \boldsymbol{x}_{n+1}, \boldsymbol{p}_n, \boldsymbol{o}_n, \bar{h}_n\big), \tag{20.3}$$

where $\boldsymbol{p}_n$ contains the declared input tuple and $\boldsymbol{o}_n$ contains X, Y, Z, J, encoded coordinates and source identifiers. The chain is

$$h_{n+1} = \mathrm{SHA256}(\mathrm{enc}\,\mathsf{P}_{n+1}), \qquad \mathsf{D}_{n+1} = (\mathsf{P}_{n+1}, h_{n+1}), \qquad h_0 = h_{\mathcal{M}}. \tag{20.4}$$

The payload excludes its own digest field while the digest is calculated. It points to a predecessor instead of recursively embedding every previous payload.

The retained projection

The earlier White Dove record operator remains

$$\Psi_{\mathrm{dove}}(\mathsf{D}, \Delta\mathsf{D}) = (\mathsf{D}, 0), \qquad \Psi_{\mathrm{dove}}^2 = \Psi_{\mathrm{dove}}. \tag{20.5}$$

This is an idempotent record-update projection. It does not set physical temperature, receptor activity or incoming sound to zero. New events append new records while previous event payloads retain their values. A reference digest detects a changed payload relative to that retained reference; it is not, by itself, a physical storage guarantee.

21 Expanding evaluation: the > self-reference

Basis: Section-21 lineage: [S3, pp. 7–9]; V4 coupling explicitly defined below.

The outgoing operation reads the model, present state and completed archive:

$$\mathcal{E}_{>} : (\mathcal{M}, \boldsymbol{x}_n, q_n, \mathcal{H}_n, \boldsymbol{p}_n) \mapsto (X_n, Y_n, Z_n, J_n, v_n, \bar{h}_n, k_{01,n}, k_{10,n}, u_n, r_n). \tag{21.1}$$

V4's numerical feature embedding is

$$v_n = \frac{Z_n^2}{1 + Z_n^2} \in [0, 1), \tag{21.2}$$

evaluated by $(Z_n/\mathrm{hypot}(1, Z_n))^2$. It is a selected dimensionless input transform, not a biological risk score. The source map itself remains unchanged.

Causal sequencing

At the start of interval n, $\bar{h}_n$ reads only v values in the already completed records $\mathcal{H}_n$. The current v_n drives the current held input but enters the archive only after the interval completes. Thus the new event affects later intervals, not the interval that created it.

The full recurrence is

$$\boldsymbol{u}_n = \mathcal{E}_{>}(\mathcal{M}, \boldsymbol{x}_n, q_n, \mathcal{H}_n, \boldsymbol{p}_n), \tag{21.3}$$

$$\boldsymbol{x}^{-}_{n+1} = \mathrm{Flow}_{\mathcal{M}}(\Delta t_n; \boldsymbol{x}_n, q_n, \boldsymbol{u}_n), \tag{21.4}$$

$$(N, q, \Theta)_{n+1} = \mathcal{T}_{\pi}(N_n, q_n, \Theta_n), \tag{21.5}$$

$$\mathsf{D}_{n+1} = \mathcal{P}_{<}\big(\mathcal{M}, \boldsymbol{p}_n, \boldsymbol{u}_n, \boldsymbol{x}^{-}_{n+1}, N_{n+1}, q_{n+1}\big), \tag{21.6}$$

$$\mathcal{H}_{n+1} = \mathcal{H}_n \| \mathsf{D}_{n+1}. \tag{21.7}$$

Continuous state is not reset by the bit event in this selected instance. The record stores the post-event bit and also the branch held during the interval.

Executable self-description

The evaluator reads a finite schema and model parameters, emits their identifiers, and then reads its own earlier output features through Equation (18.5). That is operational self-reference. The source PDF is identified as provenance, not interpreted as an arbitrary executable instruction stream. The PDF of this edition presents the rules and the results of their execution; its own final checksum is placed in the external manifest after compilation.

> The percent matrix contributes a feature; Alpha/Beta and the hourglass carry temporal state; the π event resolves a bit; commitment retains the result; evaluation reads that result into the next interval.

22 A fully declared V4 coupled instance

Basis: New V4 numerical instance. The input tuples come from [S4]; the temporal form follows [S3/S0] and explicitly selected coefficients.

The state is

$$\boldsymbol{x} = (c, a, b, m, X_{\mathrm{dwell}}, s, \vartheta), \quad \boldsymbol{x}_0 = (0.1, 0, 0, 0, 0, 0, 0), \quad q_0 = N_0 = C_0 = 0. \tag{22.1}$$

Use source example A for intervals 0–11, B for 12–23, C for 24–35 and A again for 36–47. The input angle is always $\theta = 0$. This schedule is a synthetic sequence, not a human time series.

$$k_{01,n} = \frac{1 + 0.35a_n + 0.25\bar{h}_n}{1 + 0.5m_n}\,\mathrm{s}^{-1}, \tag{22.2}$$

$$k_{10,n} = 0.6\frac{1 + 0.2b_n}{1 + 0.5m_n}\,\mathrm{s}^{-1}, \tag{22.3}$$

$$u_n = c_n + 0.4v_n + 0.2q_n + 0.2\bar{h}_n, \tag{22.4}$$

$$s_n^{\mathrm{eq}} = 0.4 + 0.25a_n + 0.2\bar{h}_n, \tag{22.5}$$

$$r_n = \left[(1 - q_n)k_{01,n} + q_n k_{10,n}\right](1 + 0.15s_n). \tag{22.6}$$

Here u_n is the held Alpha input. Set

$$\tau_\alpha = 0.2\,\mathrm{s}, \quad \tau_\beta = 4\,\mathrm{s}, \quad \chi = 0.2, \quad \eta = 0.45\,\mathrm{s}^{-1}, \quad \tau_r = 5\,\mathrm{s}. \tag{22.7}$$

The selected $k_T = 1\,\mathrm{s}^{-1}$ is a fixed reference-temperature rate. No numerical drug dose, receptor occupancy or physiological time constant is inferred from it.

Held spin response

With $\tau_\omega = 2\,\mathrm{s}$ and $\omega_\star = 1\,\mathrm{rad\,s}^{-1}$,

$$s' = s_n^{\mathrm{eq}} + (s - s_n^{\mathrm{eq}})e^{-\Delta t_n/\tau_\omega}, \tag{22.8}$$

$$\vartheta' = \vartheta + \omega_\star\left[s_n^{\mathrm{eq}}\Delta t_n + (s - s_n^{\mathrm{eq}})\tau_\omega(1 - e^{-\Delta t_n/\tau_\omega})\right]. \tag{22.9}$$

The local tensor readout uses normalized principal values 4 and 1. It is a recorded orientation diagnostic, not a solved field of measured permeability.

Each interval uses the integer divider, exact chemical update, exact Alpha/Beta exponential, exact reservoir step and exact held-spin integral. After completion the bit flips, the encoded slash is formed, and the record is appended.

Source input blocks and feature values

Interval indices	Source tuple	Z	$v = Z^2/(1 + Z^2)$
0–11	A	0.414940	0.146885414
12–23	B	577.744377	0.999997004
24–35	C	466.663622	0.999995408
36–47	A	0.414940	0.146885414

History-off comparison

The comparison keeps the same source inputs, initial state, phase convention and flow equations, but sets three archive-readout gains to zero:

$$g_{k,h} = 0, \qquad g_{u,h} = 0, \qquad g_{\omega,h} = 0. \tag{22.10}$$

In the main run those gains are 0.25, 0.2 and 0.2. Setting them to zero removes the explicit archive-readout contribution to the forward rate, Alpha input and spin target. It does not delete the archive, Clock Beta, hourglass memory, the current feature v_n, or the bit's dependence on the event count.

This is a controlled computational comparison of the selected recurrence. A difference in later interval times demonstrates that retained model output has become a dynamical input. It is not a comparison of medical treatments or a validation of the source's clinical state labels.

Event numbering and display orientation

The input block indexed by n drives the interval from t_n to t_{n+1}. Row $n + 1$ stores its resulting state, the old and new bits, and the post-event encoded slash $(-1)^{q_{n+1}} Z_n$. Source-coordinate J_n and encoded-coordinate $J_{\mathrm{enc},n+1}$ are both retained; only the former is used for the source numerical region.

Within a fixed input block, successive displayed slash directions alternate while the source tuple is unchanged. This is the deliberately defined binary orientation layer. Its geometric half-turn does not claim that the external sound field, a receptor variable or a person's condition has changed.

Named model schema

The executable schema is `klootwijk-v4-percent-source-pi-clock`. The model JSON contains all numerical coefficients, the input-block schedule, reference units, source checksums and evaluator checksum. The specified program uses Python's standard library and serializes the complete event payload in canonical key order.

23 Executed results, proofs and regression checks

Basis: New execution of the supplied `tracker_v4.py`*; full data in the companion package.*

n	t_n (s)	q_n	Source J	c_n	b_n	m_n
1	1.000000	1	6.751154	0.519004	0.029049	0.000000
2	2.618822	0	6.751154	0.636335	0.273676	0.517353
12	14.919279	0	6.751154	0.656111	1.038970	0.727762
13	15.777792	1	-15.997154	0.663174	1.083022	0.455172
24	28.035072	0	-15.997154	0.683175	1.624449	0.745193
25	28.713495	1	-8.270377	0.688889	1.620286	0.523759
36	40.338150	0	-8.270377	0.685783	1.717488	0.747174
37	40.996396	1	6.751154	0.691291	1.668842	0.531657
48	53.351134	0	6.751154	0.668761	1.273182	0.737534

The table displays selected post-event states. The source input changes on rows 13, 25 and 37. A returned bit value of zero is not a reset of the chemical, retained, occupancy, spin or history state.

Completion and self-reference comparison

Run	t_{48} (s)	$X_{\mathrm{dwell},48}$ (s)
V4	53.351134120104	35.174295014212
Archive-readout gains off	56.276188651577	35.918422178462

The completion-time difference is 2.925054531473 s. The final bit is zero and the exact stored phase coefficient is 48 in both runs. The two runs produce 48 events each. Their difference is computed from accumulated integer reference cycles, not from rounded table values. The main run's input tuples remain the source's A/B/C values even though the event times depend on the retained history.

Regression outcome

Check or quantity	Result
Main-run algebra/record checks	15 / 15 passed
History-off algebra/record checks	15 / 15 passed
Independent unit tests	21 / 21 passed
Largest population-sum residual	0
Largest tensor trace/determinant residual	2.220446×10^{-15}
Largest divider excess interval	1.025591×10^{-10} s
Canonical replay	Identical

The 15 run checks cover parity, exact phase coefficient, integer-cycle accounting, bounds, source-coordinate replay, encoded phase, tensor spectrum, source binding, record-chain linkage, divider error and deterministic replay. The 21 unit tests also exercise the root formula, filter-product sensitivity, C++ branch precedence, the distinction between overlap and determinant, and source example reproduction.

The event-time trace

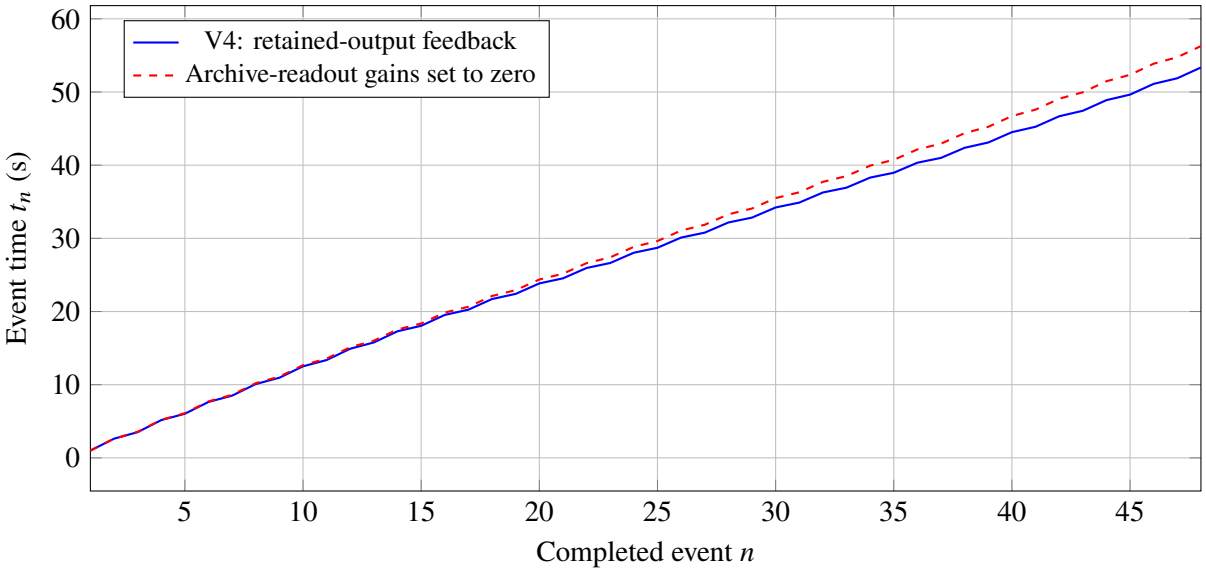


Figure 3. Executed event timing for the declared A/B/C/A input schedule. Both curves count the same 48 bit flips; they differ in the accumulated elapsed time.

Invariants that are actually proved

For nonnegative selected inputs and positive rates, the exact chemical and reservoir maps preserve their intervals. The normalized history remains a convex combination of retained features. The bit and phase obey $q_n = n \bmod 2$ and $\Theta_n = n\pi$. Orthogonal tensor transformation preserves principal values. The ceiling divider adds less than one reference period beyond the intended interval. These statements follow from the formulas and are separately checked on the executed records.

The numerical run verifies a finite instance of those mathematical operations. It supplies no sensor measurements or evidence that the source's proposed diagnostic or restorative associations are correct. All clinical-sounding strings in the event JSON are the attachment's legacy code labels, retained alongside neutral numerical region identifiers.

24 Technical disclosure: field, background and summary

Basis: Patent-style abstract, background and full disclosure: [S4, pp. 37–43].

Source title: *System and Method for Cross-Modal Reality-Gating Calibration and Trefoil Overlap Stability Tracking.*

Technical field

The source positions the proposal in human–computer interfaces, biometric signal processing, immersive rendering and spatial audio. Its concrete computational core extracts or receives numerical feature inputs, forms a Hadamard node, evaluates a coupled slash coordinate and determinant, and exposes outputs to a rendering interface.

Background, as a source position

The disclosure describes an intended gap between ordinary tracking metrics and the proposed additional gating/salience feature layer. Its categorical statements about what all existing devices lack are part of the source's claimed motivation. The attachment contains no prior-art search establishing that exclusivity, and V4 does not convert those statements into a novelty finding.

Summary of the proposed implementation

A feature-input layer provides M, G, μ, d, θ and optional s_μ or state flags. A processing layer calculates $X = \pi M$, $Y = G \odot \mu$, $Z = d \cos\theta/(XY + \epsilon)$ and $J = Xd - YZ$. The source's independent classifier variants can label the resulting numerical state. A proposed output layer uses that result to select rendering changes. V4 adds the inherited π-clock/record wrapper as an explicitly separate formal integration.

As-recorded disclosure metadata

Source field	Entry in the attachment
Inventor	Tom Klootwijk
Assignee	Holo-Sensory Systems Ltd. / Principal Tech Repository
Document reference	`PAT-NL-200678942-1007-FD`
Source classifications	G06F 3/01; A61B 5/16; H04S 7/00
Source status wording	Formal Recordation / Specification Phase

These are source metadata entries. The attachment does not provide a signed assignment, official filing receipt, granted claim, or dated priority document. This edition is a technical specification and disclosure draft; it does not assert those external legal events.

25 Drawings, signal flow and hardware contracts

Basis: Source drawing descriptions and hardware diagram: [S4, pp. 44, 48–49].

The source describes three drawings: the system architecture, the percent/Trinity arrangement, and the computation/feedback flow. Its hardware diagram is a logical signal-routing diagram. It does not include component part numbers, connector pin assignments, analog front ends or a complete electrical schematic.

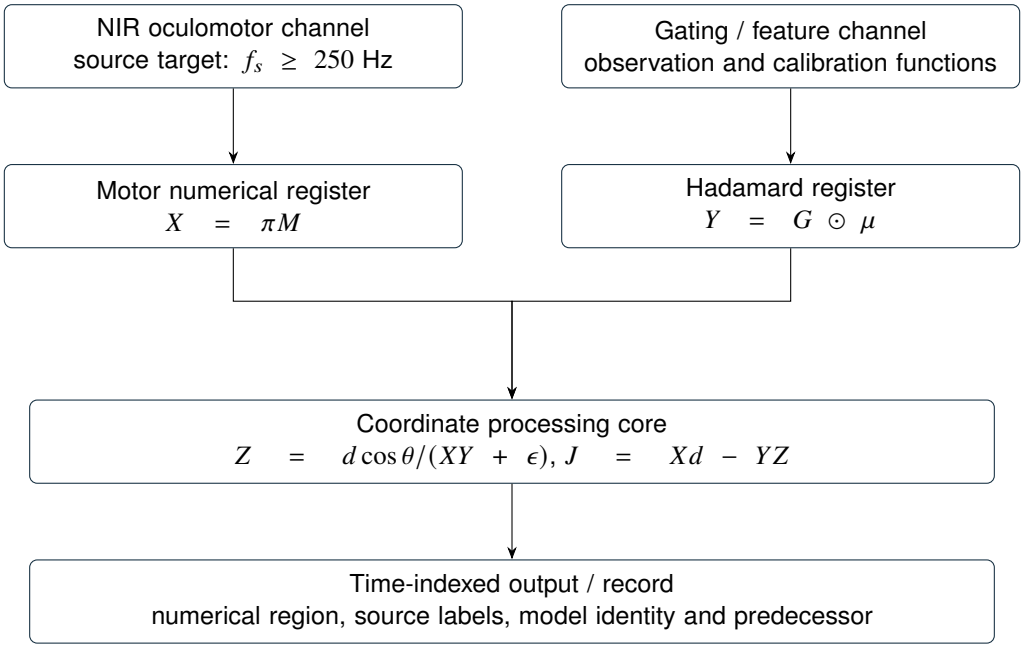


Figure 4. V4 typesetting of the source's principal data paths. The temporal record is the explicitly integrated V4 output.

Source bus names and what remains unspecified

The source names USB 3.2 Gen2 or PCIe Gen3 x1 on the eye-input path, a "Biomedical SPI Bus" on the neural-input path, DMA ingestion, vector/matrix registers, and I2S/DisplayPort on output. These names are retained as source interface suggestions, not verified interoperability or wiring claims.

At a stated 250-Hz sample rate the sample period is $1/250\,\mathrm{s} = 4\,\mathrm{ms}$, a numerical consequence of that target. It is not a measured end-to-end latency. Feature windows, averaging, synchronization, transport delays and rendering latency must be specified separately. The attachment gives no complete latency budget or calibrated live path for μ and d.

26 Six disclosure claims and their formal counterparts

Basis: Six source claims: [S4, pp. 46–47]. The entries below trace the draft rather than assert filing or clinical validity.

Claim	Source's proposed scope	Mathematical / implementation counterpart
1	Sensor array, gating/reward product, spatial audio, percentage-space core and threshold-triggered output.	Input contract, $\mathcal{P}_{\%}$, $\boldsymbol{M}_{\text{state}}$ and a proposed renderer interface. Actual calibration and actuator transfer functions are not supplied.
2	Evaluate $J = Xd - YZ$ and select an override when $J \leq 0$.	The arithmetic and comparison can be implemented exactly. The sign is a model-region predicate, not an independently established clinical finding.
3	Detect cessation of eye movements as a catatonic phase-lock and shift the audio azimuth.	The source condition refers to M near zero, but supplies no validated diagnostic observation rule or actuator-response data.
4	Inject high-pass micro-jitter into spatial-audio delay lines.	A proposed signal-processing operation; the source provides no filter coefficients, output amplitude or delay-line implementation.
5	Identify waking REM-intrusion paralysis from a small motor vector in an inactive environment and use visual jitter.	The narrative and the earlier C++ awake/REM predicates are preserved separately. Their claimed physiological equivalence is not demonstrated by the code.
6	A decreasing Hadamard product exponentially increases the slash and forces a topological threshold crossing.	The displayed law is rational and regularized; its exact derivative and determinant behavior are given in Section 11. The source claim is not substituted for that algebra.

A claim-compatible technical core

The source supports a clearly described numerical processing chain: receive declared feature values, compute a product node, compute a coupled rational coordinate, construct a 2×2 matrix, evaluate its determinant, and publish the result through a named interface. That can be disclosed independently from assertions that an output prevents, identifies or reverses a medical condition.

The more specific biological and restorative clauses require information not supplied in the attachment. V4 leaves them traceable as proposed claims rather than adding a fabricated calibration, clinical result, legal priority or prior-art conclusion.

27 Source differences and unresolved interfaces

Basis: Concordance of source prose, equations, code and figure.

Source location	Difference or missing definition	V4 treatment
pp. 21–22	G includes zero in its range but appears as $1/G$.	Reciprocal equation has domain $G > 0$; no silent cutoff is added.
pp. 22, 25	General GABA/MOR denominator versus C++ $2.5\mu + 1.5$.	Both equations are retained and their specialization stated.
pp. 23–26	Table thresholds versus ordered code predicates.	Source ranges and exact C++ precedence are separate.
pp. 26–30	Author asks for two product nodes; code explicitly gives only $G\mu$ as the right node.	Preserve author's broader description and scalar code $X = \pi M$.
pp. 29–31	Peaked graphical surface and trefoil wording lack a supplied generating embedding.	Source figure is documentary; no invented mesh or knot proof.
pp. 30–36	Overlap, C++ salience and Python determinant classifiers differ.	Three named algorithms, independent outputs.
pp. 32–33	Determinant described as proof of temporal/clinical collapse.	Derive its singularity boundary; do not infer an unspecified time law.
pp. 44–45	Sensor descriptions omit a full calibration of G, μ, d.	Explicit observation contract remains unfilled; examples are synthetic.
pp. 45–47	Falling Y is said to force negative determinant and exponential growth.	Retain rational formula; show its actual derivative and finite bound.
pp. 37–49	Patent metadata and completed-filing language are not accompanied by filing evidence.	Technical-disclosure draft with as-recorded metadata.

These are explicit source distinctions, not undisclosed replacements. V4 makes the numerical construction executable while retaining the exact points where the source's interpretation, data, or implementation does not yet specify a complete map.

28 Formal closure

Basis: Combination of the source coordinate algebra with the inherited tick and record architecture.

The completed V4 state is

$$\mathcal{S}_n = (\boldsymbol{x}_n, N_n, q_n, \Theta_n, C_n, \mathcal{H}_n, \mathcal{M}), \tag{28.1}$$

with current source input $\boldsymbol{p}_n = (M_n, G_n, \mu_n, d_n, \theta_n)$. The full update is a finite composition,

$$\boxed{\mathcal{S}_{n+1} = \mathcal{A} \circ \mathcal{P}_< \circ \mathcal{J}_\pi \circ \mathrm{Flow}_{\mathcal{M}} \circ \mathcal{E}_>(\mathcal{S}_n, \boldsymbol{p}_n).} \tag{28.2}$$

The evaluation stage computes the source coordinates and held inputs; the flow stage advances continuous variables; the event stage resolves the bit and phase endpoint; commitment creates the source-bound payload; append extends the archive.

Four exact identities

$$P_{1,n} + P_{2,n} = P_\star, \tag{28.3}$$

$$J_n = X_n d_n - \frac{Y_n d_n \cos\theta_n}{X_n Y_n + \epsilon}, \tag{28.4}$$

$$(N_{n+1} - N_n,\ q_{n+1} \oplus q_n,\ \Theta_{n+1} - \Theta_n) = (1, 1, \pi), \tag{28.5}$$

$$\mathcal{H}_{n+1} = \mathcal{H}_n \| \mathsf{D}_{n+1}. \tag{28.6}$$

The first preserves the chosen preparaat partition. The second states the exact new source index. The third states the author's tick rule. The fourth is the causal record recurrence.

What V4 delivers

It delivers a traceable source-based specification, exact determinant analysis, reproduction of the source's three numerical cases, separately preserved algorithm variants, a selected clocked integration, and a technical-disclosure map. It makes the architecture refer to its own finite rule descriptor and retained output through an executable read/compute/commit cycle.

The unresolved biological transfer functions, sensor calibrations and proposed therapeutic effects remain unresolved in the supplied source. The exact numerical construction does not need those gaps to be filled with invented information: it stands as the specified model, with its mathematical consequences and its application hypotheses distinctly identified.

Two preparaats. Two nodes and a slash. One counted event.

source input → Hadamard coordinates → π-tick → record

record → next source-driven interval

Source concordance and references

[S4] *antihistamine hatman.pdf*. Author-supplied 50-page dialogue, attached as the base of Version 4. The principal mathematical source is pp. 21–36; technical disclosure and hardware routing occupy pp. 37–49. Clinical background and interpretation hypotheses occupy pp. 1–21. The source figure reproduced here is on p. 29.

[S3] *wait I got 2 preparaats.pdf*. Author-supplied 15-page dialogue. Preparaat balance, Alpha/Beta and event rule: pp. 3–5; model-bearing commitment and causal self-reference: pp. 6–9; open-cup, stick/string, rotation and whaletail descriptions: pp. 10–15.

[S0] *trewel.pdf*. Earlier author-supplied 49-page dialogue. OTAN2 and optical lane: pp. 3–16; crosshatch and hourglass: pp. 16–23; thermal rates: pp. 23–25; dual-clock/history and recovery formulas: pp. 26–30 and 47–48.

[P] Tom Klootwijk's instruction in this conversation to formalize his architectural paradigm, add biochemical and atomic-clock equations, and make the description self-referential, with ticking interpreted as one stored-bit flip and a π phase advance; followed by requests for V3 and V4 using the respective attached sources.

[E1] Bureau International des Poids et Mesures. *The International System of Units (SI)*. Defining caesium-133 frequency and Planck constant. Consulted 16 October 2026. External metrology reference used for the retained atomic-reference channel, not for the clinical interpretation.

[E2] National Institute of Standards and Technology. *FIPS PUB 180-4, Secure Hash Standard*, August 2015. Message-digest specification. Consulted 16 October 2026. External technical reference for the selected SHA-256 record identifier.

Attribution and evidence boundaries

Tom Klootwijk, Principal Creative Technologist.
NL200678942 / 10-07-1990.

This edition preserves the author's requested architecture and source terminology. Numerical tables and regression results were newly computed from the identified V4 evaluator. The source's clinical interpretations and draft filing claims are attributed to their source passages, not presented as independent results of this computation. External references [E1–E2] identify only the metrology and digest components.

A Source, model and evaluator identities

Basis: SHA-256 values computed from the files used for this edition's execution.

Primary source PDF SHA-256

`040bb39a12614a881f42114a556a6498`
`d2ec0be0f72fb3705c4ab9fd61661c78`

Canonical model SHA-256

`4140860dfda3c6c76657ba7af65e7e32`
`38ef7996f1a1bfcc6add0334af125659`

Evaluator file SHA-256

`47c6ee4b9442d6934cb48a643ad0d8f7`
`9438f2372a7d5affba9e6635dc4028f5`

Final event digest

`75a9f8c15eaed2bd25cb809c04553467`
`e5b8ec98bb8474cfbf7c3b436b8ba1e2`

Record identity and file identity

The model identifier hashes canonical JSON serialization, not the whitespace-formatted JSON file bytes. The evaluator and source identifiers hash file bytes. The release manifest separately records byte hashes of the final PDF and all delivered files. It excludes its own content from that file list.

A completed event contains the model identifier, primary source identifier, evaluator identifier and predecessor digest. It does not contain an infinitely expanded copy of its own history. A source or parameter change produces a different identified execution; a matching canonical replay reproduces the same event sequence in the stated arithmetic environment.

Execution commands

```
python tracker_v4.py --model model_v4.json --out reproduced
python -m unittest -v test_tracker_v4
```

Run these commands from the `package` subdirectory. The program uses Python 3.10 or later and the standard library. The published run used Python 3.13.5. Both the main and history-off run passed all 15 record/algebra checks; the separate suite passed 21 unit tests.

B Implementation and data dictionary

Basis: Delivered reference implementation and serialized execution records.

Record field group	Meaning
`inputs`	Declared source-example tuple used during the interval.
`x,y,z,determinant`	Original source coordinates and matrix index.
`region`	Neutral numerical region from Equation (12.2).
`source_python_label`	Literal source program string, retained as provenance.
`source_overlap_flag`	Separate coordinate-overlap predicate.
`q_before,q,n`	Old interval bit, complemented endpoint bit and event number.
`theta_over_pi`	Exact integer coefficient of the unwrapped event phase.
`z_encoded, determinant_encoded`	New V4 binary-orientation view, distinct from source coordinates.
`atomic_cycles, divider_cycles`	Cumulative and interval integer reference cycles.
`t,dt`	Endpoint and interval durations in seconds.
`v,history_readout`	Current embedded feature and readout of prior recorded features.
`k01,k10,rate,held_input`	Held kinetic and timing inputs.
`c,a,b,m,dwell`	Resulting chemical, dual-clock and hourglass state.
`spin,rotor_angle,kxx,kxy,kyy`	Selected mechanical-state and normalized local tensor readout.
`source_id,model_id,implementation_id`	Identifies the source, model descriptor and producing evaluator.
`previous_digest,digest`	Event-chain identity fields.

Offline exploration

The supplied HTML explorer evaluates the unchanged source scalar formula as its controls move. Its tick button complements one displayed bit and reverses the encoded slash; the source angle and source determinant remain visible separately. It uses no external libraries, clinical sensor feed or audio/visual actuator. The full temporal integration and canonical record replay are supplied by the Python evaluator.

The editable LaTeX source reads the delivered CSV records for its plots. The source table fragments contain the actual published outputs. Parameter edits create a new numerical execution and require regeneration of the corresponding table fragments before republishing the PDF.